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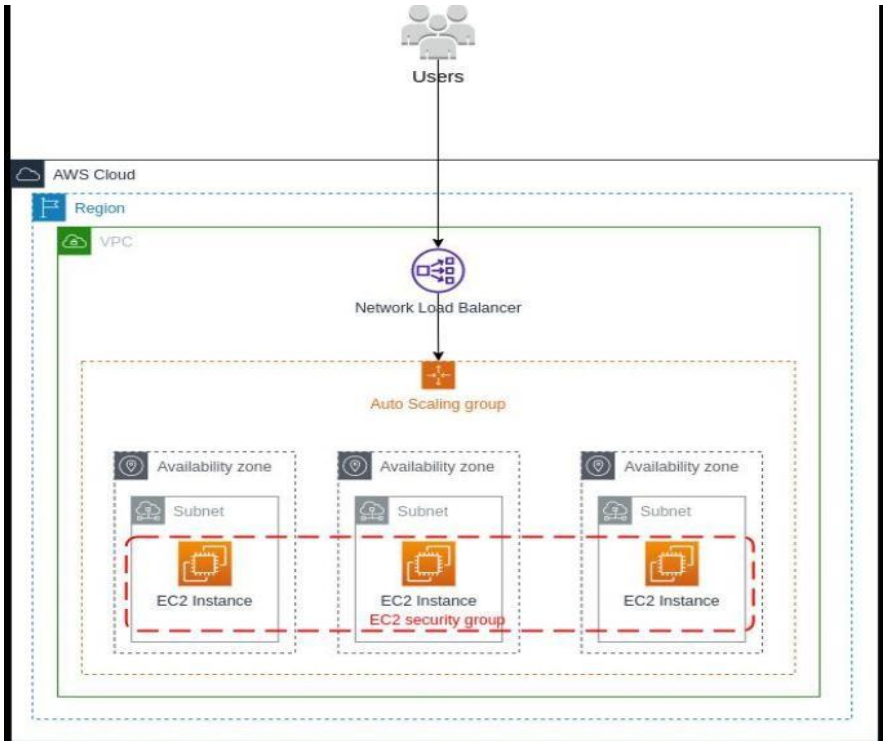
Project: 2

Flux Scale: High-performance scalable web app infrastructure



Project Overview (Simple Explanation)

Flux Scale is as a lable web application architecture built using AWS that can handle heavy traffic with low latency. It automatically adjusts resources based on demand and ensures high availability using load balancing and scaling techniques.

AWS Services Used Why

1. AmazonEC2

Why used:
EC2 instances act as servers where your web application runs.

Script/Explanation:

- Launch multiple EC2 instances
- Deploy your web app (HTML, Java, Python, etc.)
- These instances handle user requests

–Think: “Actual machines running your app”

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2. Network Load Balancer

Why used:
To distribute incoming traffic efficiently with **ultra-low latency**.

Script / Explanation:

- Accepts user requests
- Distributes traffic across EC2 instances
- Works at Layer 4 (TCP/UDP)→faster than ALB

Think: “Traffic controller for high-speed requests”

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3. 🧪 Amazon EC2 Auto Scaling

Why used:
Automatically increases or decreases EC2 instances based on traffic.

Script/Explanation:

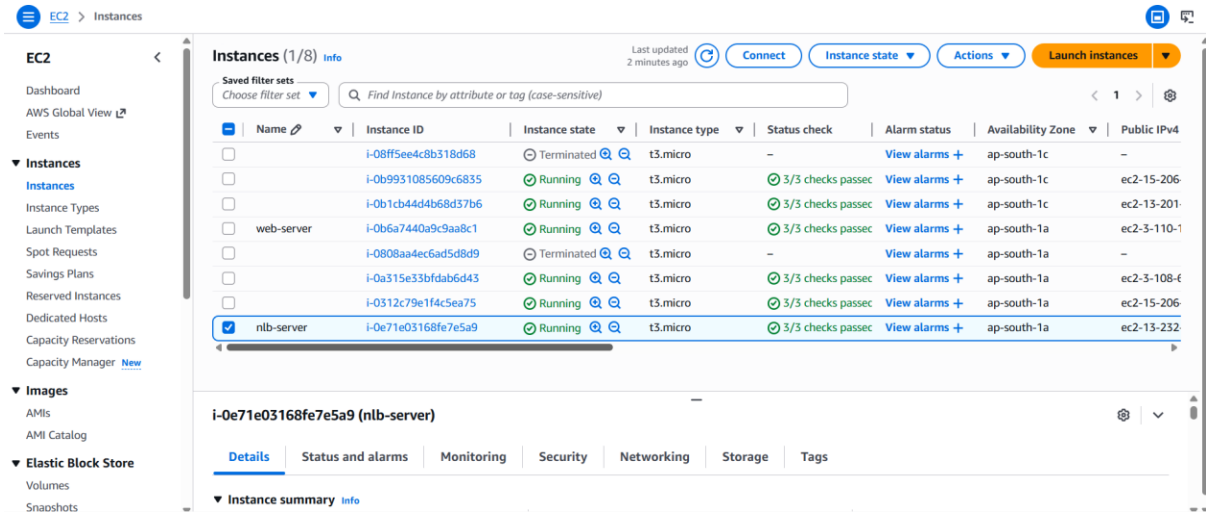
- Monitors CPU usage/traffic
- Adds instances when load increases
- Removes instances when load decreases

Think: “Auto expansion and shrink system”

🟦 How This Project Works (Flow)

1. User sends request
2. Network Load Balancer receives request
3. NLB forwards request to EC2 instances
4. Auto Scaling ensures enough instances are running
5. User gets fast and reliable response

🧪 Summary

Flux Scale is a highly scalable and low-latency web application architecture using EC2, Network Load Balancer, and Auto Scaling. It ensures efficient traffic handling, automatic scaling based on demand, and high availability, making it suitable for high-performance applications like gaming, streaming, or real-time systems.

NLB Server Running